A QUASI-COMPACT IMMERSION OF ALGEBRAIC SPACES FACTORS AS AN OPEN IMMERSION FOLLOWED BY A CLOSED IMMERSION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a descent along a presentation; the flat base change property of scheme theoretic images is the step that makes the descent possible.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Scheme theoretic images of quasi-compact morphisms commute with flat base change. Morphisms, Lemma morphisms-lemma-flat-base-change-scheme-theoretic-image, used at proof lines 2386-2391. Gives the equality s^{-1}Tbar = t^{-1}Tbar needed for descent.
- Descending subspaces through a presentation of an algebraic space. Properties of Spaces, Lemma spaces-properties-lemma-subspaces-presentation, used at proof lines 2392-2395. Produces the closed subspace Zbar of X with pullback Tbar.
- The scheme case: a quasi-compact immersion is an open immersion onto its scheme theoretic image. Morphisms, Lemma morphisms-lemma-quasi-compact-immersion, used at proof lines 2396-2398. Makes T -> Tbar an open immersion, hence Z -> Zbar as well.
- Surjective etale presentations of algebraic spaces. used at proof lines 2384-2386. Supplies U, R and the projections used throughout.

Source reference. Original lemma 081I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10218–10218. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01RG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2301–2301. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 07TW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9951–9951. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $Z \to X$ be an immersion of algebraic spaces over S. Assume $Z \to X$ is quasi-compact. There exists a factorization $Z \to \overline{Z} \to X$ where $Z \to \overline{Z}$ is an open immersion and $\overline{Z} \to X$ is a closed immersion.*

Proof. Let U be a scheme and let $U \to X$ be surjective étale. As usual denote $R = U \times_X U$ with projections $s, t : R \to U$. Set $T = Z \times_U X$. Let $\overline{T} \subset U$ be the scheme theoretic image of $T \to U$. Note that $s^{-1}\overline{T} = t^{-1}\overline{T}$ as taking scheme theoretic images of quasi-compact morphisms commute with flat base change, see Morphisms, Lemma 081I. Hence we obtain a closed subspace $\overline{Z} \subset X$ whose pullback to U is $\overline{T}$, see Properties of Spaces, Lemma 07TW. By Morphisms, Lemma 01RG the morphism $T \to \overline{T}$ is an open immersion. It follows that $Z \to \overline{Z}$ is an open immersion and we win. □